

AbleSpace Product Understanding: Take Data Workflow & UX Analysis

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Assessment: AbleSpace Full Stack Developer Technical Assessment (Part 2 — Product Understanding)

Target Platform: AbleSpace.io (<https://ablespace.io/>)

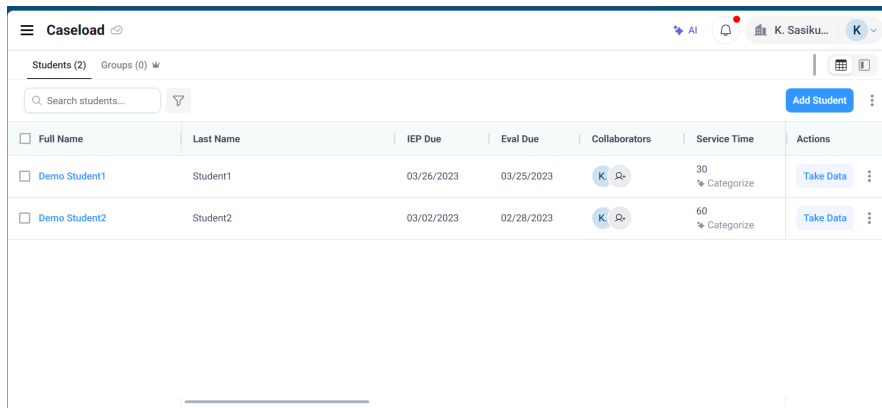
1. Executive Summary

AbleSpace is built specifically for Special Education (SPED) teachers, Speech-Language Pathologists (SLPs), and therapists who manage high caseloads of students with Individualized Education Programs (IEPs). Tracking real-time behavioral and academic progress in dynamic, fast-paced classroom environments is central to their daily workflow. This document reviews the end-to-end "Take Data" workflow from the Caseload view, analyzes friction points in the user journey, and proposes concrete UI/UX and functional enhancements aimed at reducing cognitive load and saving educators valuable time.

2. Workflow Overview: How to "Take Data"

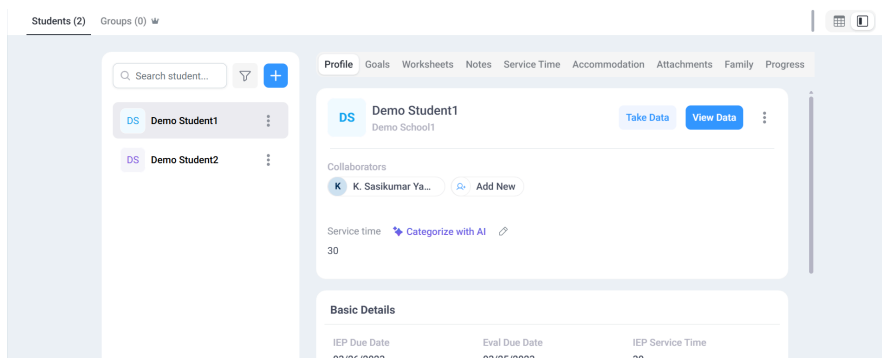
Step 1: Accessing the Caseload

- Action: The educator logs into the platform and navigates to the Caseload tab located in the left navigation sidebar.
- Purpose: The Caseload dashboard acts as the primary command center, listing all assigned students such as Demo Student1 and Demo Student2. It provides immediate visibility into IEP/Eval due dates and service time tracking, with direct "Take Data" and "Add Student" buttons available on the Caseload table.



Step 2: Selecting a Student

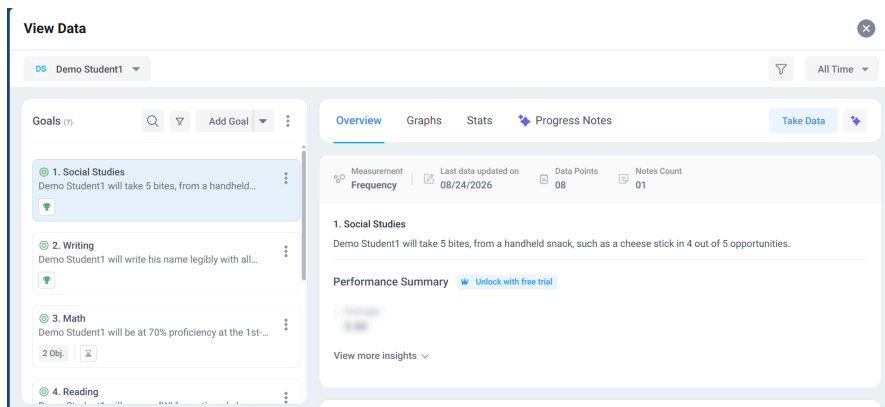
- Action: The educator selects a specific student card or row from the caseload list to open the student's profile.
- Purpose: This reveals a comprehensive student profile view featuring specific tabs for Goals, Worksheets, Notes, Service Time, Accommodation, Attachments, Family, and Progress. Educators can also view collaborators and access quick-action buttons for efficient management.



Step 3: Taking Data & Goal Overview

- Action: The user clicks the prominent "Take Data" button to launch the live data collection interface.
- Purpose: Educators can view the student's full IEP goals list (e.g., Social Studies, Writing, Math, Reading) and monitor performance through frequency measurement metric cards like Data Points and Notes Count before launching the "Take Data" interface.
 - Accuracy / Opportunity Tracking: Correct vs. Incorrect trials (e.g., 8/10 trials = 80% accuracy).

- Prompt Hierarchy Levels: Independent, Visual, Verbal, Gestural, or Physical prompt usage.
- Behavioral Frequency & Duration: Frequency tally counters, interval recording, or duration timers.
- Session Notes: Contextual notes regarding student demeanor, environment, or accommodations used.



Step 4: Saving & Reviewing Progress

- Action: The educator clicks "Save / Complete Session".
- Purpose: Historical trial data is logged under the Stats tab in a detailed table showing session timestamps, recorded frequency counts, session notes, and audit history, automatically updating longitudinal progress charts.

Session/Trial	Phase	Edited By	Frequency	Notes
01/01/2023 11:25 AM	-	K	1	
01/06/2023 12:26 PM	-	K	2	1. Demo Student hitting other s See more
01/22/2023 12:43 PM	-	K	3	
02/12/2023 01:23 PM	-	K	3	

3. UX/UI & Functionality Improvements

Improvement 1: Quick-Log & Keyboard Shortcuts (Reducing Click Depth)

- Observation: Logging a single data trial requires multiple navigation steps (Caseload → Student Profile → Goal → Take Data → Submit). In a busy classroom where a teacher manages multiple students simultaneously, this click overhead causes distraction and delayed recording.
- Proposed Solution:
 - Add a 1-click "Quick Log" drawer or modal directly accessible from the main Caseload list.
 - Introduce keyboard shortcuts for desktop users (e.g., Key 1 / + for Correct, Key 2 / - for Incorrect, Enter to save trial) allowing rapid, eyes-free data collection during live instruction.

Improvement 2: Multi-Student / Group Session Mode (Bulk Data Entry)

- Observation: Teachers and therapists frequently conduct group therapy or small-group reading/math sessions with 3 to 5 students at once. Switching back and forth between individual student profiles to record simultaneous trials is cumbersome.
- Proposed Solution:
 - Implement a "Group Session Mode" allowing educators to select multiple students from the Caseload and display a split/grid card view.
 - Educators can tap data points for Student A and Student B on the same screen without navigating away.

Improvement 3: Visual Contrast & Accessibility Compliance (WCAG AA)

- Observation: In classroom lighting conditions or on mobile tablets, subtle gray-on-white text and low-contrast metric selector buttons can be difficult to distinguish quickly.

- Proposed Solution:
 - Increase text contrast ratios to meet WCAG AA standards (minimum 4.5:1 for normal text).
 - Use distinct, color-coded status badges with high-contrast borders and clear typographic hierarchy for prompt levels (e.g., Independent vs. Prompted).

Improvement 4: Multi-Sensory Feedback on Save (Micro-Animations & Audio/Haptic Confirmation)

- Observation: When saving a trial or completing a session, the toast confirmation is small and easily missed, leaving teachers uncertain if their data synced properly.
- Proposed Solution:
 - Incorporate immediate micro-animations (such as a smooth checkmark transition on the save button) along with optional gentle haptic feedback on mobile devices and an optional subtle audio chime.

Improvement 5: Offline Mode with Auto-Sync Caching

- Observation: Schools often suffer from spotty Wi-Fi coverage across sensory rooms, playgrounds, or temporary classrooms. Losing internet connection during data collection risks losing session logs.
 - Proposed Solution:
 - Implement local client-side storage (IndexedDB / LocalStorage) with service worker caching so educators can take data completely offline.
 - Automatically sync stored logs to the backend once a stable internet connection is re-established, showing a clear offline status indicator.
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4. Value Proposition Summary for Special Education

By streamlining data entry speed, introducing group session tracking, and reinforcing reliability through offline sync and high-contrast UI, AbleSpace

significantly diminishes the administrative burden on special educators. This enables teachers to dedicate more attention to direct student engagement while maintaining compliance and audit-ready IEP documentation.
